

Foundation (Jalapeno)

- Q1.** What is the main purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To compare the dispersion of two datasets
(C) To identify outliers in a dataset
(D) To show the frequency of each data point
(E) To calculate the range of a dataset
- Q2.** Which part of a box-and-whisker plot represents the interquartile range?
(A) The whiskers
(B) The box
(C) The median line
(D) The outliers
(E) The axes
- Q3.** What is the Q_1 in a box-and-whisker plot?
(A) The median of the lower half of the data
(B) The median of the upper half of the data
(C) The mean of the entire dataset
(D) The range of the entire dataset
(E) The mode of the entire dataset
- Q4.** Which of the following is an advantage of using a box-and-whisker plot?
(A) It can display only a small dataset
(B) It can be used to compare multiple datasets
(C) It is difficult to read and understand
(D) It does not show the distribution of the data
(E) It is sensitive to outliers
- Q5.** What is the term for data points that are far away from the rest of the data?
(A) Outliers
(B) Inliers
(C) Median
(D) Mode
(E) Range
- Q6.** A box-and-whisker plot can be used to identify $\frac{1}{4}$ of the data that is below the:
(A) Mean
(B) Median
(C) Mode
(D) Lower quartile
(E) Upper quartile
- Q7.** The interquartile range is calculated by subtracting the:
(A) Lower quartile from the upper quartile
(B) Upper quartile from the lower quartile
(C) Mean from the median
(D) Mode from the median
(E) Range from the median
- Q8.** What does a box-and-whisker plot display about the dispersion of a dataset?
(A) The mean and standard deviation
(B) The median and range
(C) The quartiles and interquartile range
(D) The mode and frequency
(E) The variance and coefficient of variation

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Identify the median, quartiles, and interquartile range.
- Q10.** Describe the differences between a box-and-whisker plot and a histogram. How do they display the dispersion of a dataset?

Chef's Tips

- Tip 1: Make sure to check for any mistakes in the calculations.
- Tip 2: Provide clear and concise answers to the free-response questions.
- Tip 3: Use the space provided to show all work and explain the reasoning.